

9. This problem has to do with odds. 胜率

- (a) On average, what fraction of people with an odds of 0.37 of defaulting on their credit card payment will in fact default?
- (b) Suppose that an individual has a 16% chance of defaulting on her credit card payment. What are the odds that she will default?

$$(a) \text{ odd} = \frac{p}{1-p} = \frac{\text{成功}}{\text{失败}}$$

$$p = \frac{\text{odd}}{1+\text{odd}} = \frac{0.37}{1+0.37} = 0.27 \quad \therefore 27\%$$

$$(b) \quad p = 16\% \quad \text{odd} = \frac{0.16}{1-0.16} = \frac{0.16}{0.84} \approx 0.19 \quad \therefore 19\%$$